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Finance and Investments club

Corporate Finance Primer

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FOREWORD

Greetings!

The Finance and Investment Club of IIM Calcutta is excited to present the first edition of Corporate Finance Primer.

The motive of this primer is to explain important concepts asked in corporate finance interviews. Other relevant corporate finance topics like CFRA and Valuation concepts can be referred to from the Investment Banking Primer.

Along with the primers, we would advise looking at the Finance Interview Question Bank and other materials shared by the club for preparation. Lastly, reach out to as many corporate finance interns as you can to understand their interview and internship experiences.

We hope that this primer helps in bringing you one step closer to landing your dream internship. Feel free to reach out to us for any queries/suggestions.

We wish you all the very best!

Finance and Investments Club

IIM Calcutta

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FINANCIAL RATIOS

What are Financial Ratios?

Financial ratios are mathematical comparisons of financial statement accounts or categories. These relationships between the financial statement accounts help investors, creditors, and internal company management understand how well a business is performing and of areas needing improvement.

Ratios are just a raw computation of financial position and performance. In a sense, financial ratios don't take into consideration the size of a company or the industry.

Financial ratios are grouped into the following categories:

- Liquidity ratios
- Solvency ratios
- Efficiency ratios
- Profitability ratios
- Market value ratios

Liquidity Ratios

Liquidity of a company depends on its ability to convert assets into cash or to generate cash through operations to meet short term obligations.

Some important liquidity ratios are as follows –

1. **Current Ratio**: The current ratio measures a company's ability to pay off its short-term liabilities with current assets. Higher the current ratio, greater is the assurance that current liabilities will be met. A thumb rule of 2 is widely used as an optimal ratio.

Current ratio = Current assets / Current liabilities

2. **Quick Ratio/Acid test Ratio**: It measures a company's ability to pay off short-term liabilities with quick assets. It is a more stringent test of liquidity. It includes only those current assets which can be readily converted into cash. Inventories are excluded because they are least liquid and prepaid expenses are excluded because they will not get converted into cash.

Acid-test ratio = (Current assets – Inventories – Prepaid Expenses) / Current liabilities

3. **Cash Ratio**: The cash ratio measures a company's ability to pay off short-term liabilities with cash and cash equivalents. It is more stringent in comparison to the quick ratio. It emphasises that ultimately cash is required to settle current liabilities.

Cash ratio = Cash and Cash equivalents + Marketable Securities/ Current Liabilities

Note: A higher current ratio is more favourable than a lower current ratio because it shows the company can more easily make current debt payments. But high current ratio also shows underutilization of resources which could have been used to increase profitability of the company by investing them elsewhere.

Solvency Ratios

Solvency refers to the long-term viability of the company. Ratios related to capital structure and earnings coverage are used to measure solvency. It measures how leveraged the company is and how well is it placed with respect to its debt repayments capacity.

Some important solvency ratios are as follows -

1. **Debt to Total Assets Ratio**: It measures the relative amount of a company's assets that are provided from debt. Total liabilities include both current and non-current liabilities.

   Debt ratio = Total liabilities / Total assets

2. **Debt to Equity Ratio**: It calculates the weight of total debt and financial liabilities against shareholders' equity.

   Debt to equity ratio = Total liabilities / Shareholder's equity

3. **Interest Coverage Ratio**: It shows how easily a company can pay its interest expenses on the assumed debt. Higher interest coverage ratios imply the greater ability of the firm to pay off its interests. If Interest coverage is less than 1, then EBIT is not sufficient to pay off interest, which implies finding other ways to arrange funds.

   Interest coverage ratio = EBIT/ Interest expenses

4. **Debt Service Coverage Ratio**: It reveals how easily a company can pay its debt obligations. Total debt service includes Principal payment, Interest payment and Lease payment.

   Debt service coverage ratio = Operating income / Total debt service

Turnover Ratios

The turnover ratio is also known as **activity ratio**. This type of ratio indicates the efficiency with which an enterprise's resources are utilized.

Some important turnover/activity ratios are as follows -

1. **Inventory Turnover Ratio**: It measures how many times a company's inventory is sold and replaced over a given period. It is a measure of how efficiently a company can control its merchandise, so it is important to have a high turnover ratio.

   Inventory turnover ratio = Cost of goods sold / Average inventory

   Days sales in inventory = 365 days / Inventory turnover ratio

2. **Accounts Receivables turnover ratio**: It measures a business' ability to efficiently collect its receivables. Higher ratios mean that companies are collecting their receivables more frequently throughout the year. The reason net credit sales are used is because only credit sales establish a receivable.

Receivables turnover ratio = Net credit sales / Average accounts receivable

Days Accounts Receivables = 365/ Receivables turnover Ratio

3. **Accounts Payables turnover ratio**: It measures a business' ability to efficiently pay its receivables. Lower ratios mean that companies have more time to pay their payables throughout the year.

Payables turnover ratio = Net credit Purchases / Average accounts payable

Days Accounts Payables = 365/ Payables turnover Ratio

4. **Asset Turnover Ratio**: It measures a company's ability to generate sales from its assets. The total asset turnover ratio calculates net sales as a percentage of assets to show how many sales are generated from each dollar of company assets.

Asset turnover ratio = Net sales / Average total assets

5. **Cash Conversion Cycle**: The cash conversion cycle is a cash flow calculation that attempts to measure the time it takes a company to convert its investment in inventory and other resource inputs into cash. In other words, the cash conversion cycle calculation measures how long cash is tied up in inventory before the inventory is sold and cash is collected from customers.

Cash Conversion cycle = Days sales in inventory + Days Accounts Receivables - Days Accounts Payables

Profitability Ratios

Profitability ratios measure a company's ability to generate income relative to revenue, balance sheet assets, operating costs, and equity.

Common profitability financial ratios include the following -

1. **Gross Margin Ratio**: It compares the gross profit of a company to its net sales to show how much profit a company makes after paying its cost of goods sold. It basically shows % amount of money from product sales left over after all of the direct costs associated with manufacturing the product has been paid.

Gross margin ratio = Gross profit / Net sales

2. **Operating Margin Ratio**: It compares the operating income of a company to its net sales to determine operating efficiency. Many times, operating income is classified as earnings before interest and taxes. If companies can make enough money from their operations to support the business, the company is usually considered more stable.

Operating margin ratio = Operating income / Net sales

3. **Return on Assets Ratio**: The return on assets ratio measures how effectively a company can earn a return on its investment in assets. In other words, ROA shows how efficiently a company can convert the money used to purchase assets into net income or profits.

Return on assets ratio = Net income / Total assets

4. **Return on Equity Ratio**: It measures how efficiently a company is using its equity to generate profit. In other words, the return on equity ratio shows how much profit each dollar of common stockholders' equity generates.

Return on equity ratio = Net income / Shareholder's equity

Market Value Ratios

Market value ratios are used to evaluate the share price of a company's stock. Common market value ratios include the following:

1. **Book Value per share Ratio**: It calculates the per-share value of a company based on equity available to common shareholders divided by the number of outstanding shares. This figure represents the minimum value of a company's equity and measures the book value of a firm on a per-share basis.

Book value per share ratio = (Shareholder's equity – Preferred equity) / Total common shares outstanding

2. **Dividend Yield Ratio**: It measures the amount of cash dividends distributed to common shareholders relative to the market value per share. The dividend yield is used by investors to show how their investment in stock is generating cash flows in the form of dividends.

Dividend yield ratio = Dividend per share / Share price

3. **Earnings per share Ratio**: It measures the amount of net income earned per share of stock outstanding. In other words, this is the amount of money each share of stock would receive if profits were distributed to the outstanding shares at the end of the year.

Earnings per share ratio = Net income- Preferred Dividends / Total shares outstanding

4. **Price-Earnings Ratio**: It calculates the market value of a stock relative to its earnings by comparing the market price per share by the earnings per share. In other words, the price earnings ratio shows what the market is willing to pay for a stock based on its current earnings.

Price-earnings ratio = Share price / Earnings per share

WORKING CAPITAL MANAGEMENT

Working capital management refers to the set of activities performed by a company to make sure it has enough resources for day-to-day operating expenses while keeping resources invested in a productive way.

What is Working Capital?

Working capital is the difference between a company's current assets and its current liabilities.

- Current assets include cash, accounts receivable, and inventories.
- Current liabilities include accounts payable, short-term borrowings, and accrued liabilities.

Some approaches may subtract cash from current assets and financial debt from current liabilities.

Why Working Capital Management is Important?

- Ensuring that the company possesses appropriate resources for its daily activities means protecting the company's existence and ensuring it can keep operating as a going concern.
- Scarce availability of cash, uncontrolled commercial credit policies, or limited access to short-term financing can lead to the need for restructuring, asset sales, and even liquidation of the company.

Managing Liquidity

- Properly managing liquidity ensures that the company possesses enough cash resources for its ordinary business needs and unexpected needs of a reasonable amount. It's also important because it affects a company's creditworthiness, which can contribute to determining a business's success or failure.
- The lower a company's liquidity, the more likely it is going to face financial distress, other conditions being equal. However, too much cash parked in low- or non-earning assets may reflect a poor allocation of resources.
- Proper liquidity management is manifested at an appropriate level of cash and/or in the ability of an organization to quickly and efficiently generate cash resources to finance its business needs.

Managing Accounts Receivables

- A company should grant its customers the proper flexibility or level of commercial credit while making sure that the right amounts of cash flow in via operations.
- A company will determine the credit terms to offer based on the financial strength of the customer, the industry's policies, and the competitors' actual policies.
- Credit terms can be ordinary, which means the customer generally is given a set number of days to pay the invoice (generally between 30 and 90). The company's policies and manager's discretion can determine whether different terms are necessary, such as cash before delivery, cash on delivery, bill-to-bill, or periodic billing.

Managing Inventory

- Inventory management aims to make sure that the company keeps an adequate level of inventory to deal with ordinary operations and fluctuations in demand without investing too much capital in the asset.
- An excessive level of inventory means that an excessive amount of capital is tied to it. It also increases the risk of unsold inventory and potential obsolescence eroding the value of inventory.
- A shortage of inventory should also be avoided, as it would determine lost sales for the company.

Managing Short-Term Debt

- Like liquidity management, managing short-term financing should also focus on making sure that the company possesses enough liquidity to finance short-term operations without taking on excessive risk.
- The proper management of short-term financing involves the selection of the right financing instruments and the sizing of the funds accessed via each instrument.
- Popular sources of financing include regular credit lines, uncommitted lines, revolving credit agreements, collateralized loans, discounted receivables, and factoring.
- A company should ensure there will be enough access to liquidity to deal with peak cash needs. For example, a company can set up a revolving credit agreement well above ordinary needs to deal with unexpected cash needs.

Managing Accounts Payable

- Accounts payable arises from trade credit granted by a company's suppliers, mostly as part of the normal operations. The right balance between early payments and commercial debt should be achieved.
- Early payments may unnecessarily reduce the liquidity available, which can be put to use in more productive ways.
- Late payments may erode the company's reputation and commercial relationships, while a high level of commercial debt could reduce its creditworthiness.

Is a negative working capital always a bad sign?

- Most of the time, this is not considered as a good sign, but there are cases where negative working capital is good for the organization.
- Sometimes it means that the company can generate the cash so quickly that it gets time in between to pay off its suppliers and creditors. So basically, the company is using the suppliers' money to run its day-to-day operations. Generally, firms who are dealing with cash-only business enjoy high turnover with negative working capital. Such firms don't supply goods on credit and constantly increase their sales. Online retailers, discount retailers, grocery stores, restaurants and telecom companies are expected to have negative working capital
- Though this means a good idea, having the negative working capital to its advantage is not everyone's cup of tea. The companies which deal with cash only businesses or where the receivables time is too short often have negative working capital.

LEVERAGE

Leverage, in the sense we use it here, refers to the amount of fixed costs a firm has. These fixed costs may be fixed operating expenses, such as building or equipment leases, or fixed financing costs, such as interest payments on debt. Greater leverage leads to greater variability of the firm's after-tax operating earnings and net income. A given change in sales will lead to a greater change in operating earnings when the firm employs operating leverage; a given change in operating earnings will lead to a greater change in net income when the firm employs financial leverage.

- **Business risk** refers to the uncertainty about operating earnings (EBIT) and results from variability in sales and expenses. Business risk is magnified by operating leverage.
- **Financial risk** refers to the additional variability of EPS compared to EBIT. Financial risk increases with greater use of fixed cost financing (debt) in a company's capital structure.

Profitability Statement	
Sales	Xxx
Less: Variable Cost	(xxx)
Contribution	Xxx
Less: Fixed Cost	(xxx)
Operating Profit/EBIT	Xxx
Less: Interest	(xxx)
EBT	Xxx
Less: Tax	(xxx)
Profit after Tax	Xxx
Less: Preference Dividend	(xxx)
Net Earnings available to Equity Shareholders (X)	**Xxx**
No. of Equity Shares (N)	**Xxx**
EPS (X/N)	**Xxx**

The **degree of operating leverage (DOL)** is defined as the percentage change in operating income (EBIT) that results from a given percentage change in sales:

DOL = percentage change in EBIT/ percentage change in sales

To calculate a firm's DOL for a particular level of unit sales -

DOL = Contribution/ EBIT

The **degree of financial leverage (DFL)** is interpreted as the ratio of the percentage change in net income (or EPS) to the percentage change in EBIT:

DFL = percentage change in EPS/ percentage change in EBIT

For a particular level of operating earnings, DFL is calculated as:

DFL = EBIT/ EBIT - Interest

The **degree of total leverage (DTL)** combines the degree of operating leverage and financial leverage. DTL measures the sensitivity of EPS to change in sales.

DTL is computed as:

DTL = DOL x DFL

= percentage change in EPS/ percentage change in sales

Or

Contribution/ EBT

CAPITAL STRUCTURE

Both debt and equity can be found on the balance sheet. Company assets, also listed on the balance sheet, are purchased with this debt and equity. Capital structure can be a mixture of a company's long-term debt, short-term debt, common stock, and preferred stock. A company's proportion of short-term debt versus long-term debt is considered when analysing its capital structure.

When analysts refer to capital structure, they are most likely referring to a firm's debt-to-equity (D/E) ratio, which provides insight into how risky a company's borrowing practices are. Usually, a company that is heavily financed by debt has a more aggressive capital structure and therefore poses a greater risk to investors. This risk, however, may be the primary source of the firm's growth.

- **Debt** is one of the two main ways a company can raise money in the capital markets. Companies benefit from debt because of its tax advantages; interest payments made as a result of borrowing funds may be tax-deductible. Debt also allows a company or business to retain ownership, unlike equity. Additionally, in times of low interest rates, debt is abundant and easy to access.
- **Equity** allows outside investors to take partial ownership in the company. Equity is more expensive than debt, especially when interest rates are low. However, unlike debt, equity does not need to be paid back. This is a benefit to the company in the case of declining earnings. On the other hand, equity represents a claim by the owner on the future earnings of the company.

Capital structure refers to the proportion of equity vs. debt financing that a firm utilizes to carry out its operations and grow. Managers need to weigh the costs and benefits of raising each type of capital along with their ability to raise either. Equity capital involves diluting some of the company ownership and voting rights, but comes with fewer obligations to investors in terms of repayment. Debt tends to be cheaper capital (plus it has tax advantages), but comes with serious responsibilities in terms of repaying interest and principal, which can lead to default or bankruptcy if not carried through. Firms in different industries will use capital structures better-suited to their type of business. Capital-intensive industries like auto manufacturing may utilize more debt, while labour-intensive or service-oriented firms like software companies may prioritize equity.

Optimal capital structure

The optimal capital structure of a firm is often defined as the proportion of debt and equity that results in the lowest weighted average cost of capital (WACC) for the firm.

$$WACC = (E/V \times Re) + ((D/V \times Rd) \times (1 - T))$$

Where:
E = market value of the firm's equity (market cap)
D = market value of the firm's debt
V = total value of capital (equity plus debt)
Re = cost of equity (required rate of return)
Rd = cost of debt (yield to maturity on existing debt)
T = tax rate

In order to optimize the structure, a firm can issue either more debt or equity. The new capital that's acquired may be used to invest in new assets or may be used to repurchase debt/equity that's currently outstanding, as a form of recapitalization.

Capital Structure	Risk	Return	Ownership	Performance
Debt	High Risk	Low Return Interest + Capital back	No ownership Rights	Temporal
Equity	Low Risk	High Return Dividend + Capital Growth	Ownership Rights – Voting Rights	Permanent

Debt investors take less risk because they have the first claim on the assets of the business in the event of bankruptcy. For this reason, they accept a lower rate of return and, thus, the firm has a lower cost of capital when it issues debt compared to equity.

Equity investors take more risk, as they only receive the residual value after debt investors have been repaid. In exchange for this risk, investors expect a higher rate of return and, therefore, the implied cost of equity is greater than that of debt.

BUDGETING

What is a budget?

A budget is an estimation of revenue and expenses over a specified future period of time and is usually compiled and re-evaluated on a periodic basis. Budgets can be made for a person, a group of people, a business, a government, or just about anything else that makes and spends money.

It explains the company's objectives and the course of action it will choose to achieve its goals in detail. Also, it mentions the controls to be put in place for achieving its successful implementation.

The budgeting process is the process of putting a budget in place. This process involves planning and forecasting, implementing, monitoring and controlling, and finally evaluating the performance of the budget.

There are two main approaches to the budgeting process. These are:

Top-down approach

This budgeting process involves preparing the budget by the company's senior management based on the company's objectives. The departmental managers are assigned the responsibility for its successful implementation. Every department can opt to create its own budget based on the company's broader budget allocation and goals.

This approach's advantage is that the lower management saves a lot of time and gets a readymade budget to be followed. They hardly participate in the preparation of the central budget. The senior managers' experience, coupled with past-performance figures, comes in handy in such budgeting processes.

Bottom-up approach

This budgeting process starts at the departmental level and moves up to higher levels. Every department within the company is required to prepare plans for its proposed activities for the next budget period and estimate the costs it will incur. These individual budgets are combined to create a bigger all-inclusive budget.

The budgeting process with this approach can be lengthy and time-consuming. However, employees and managers are more motivated to achieve the budget goals since they have prepared it. They have the complete knowledge of what the budget actually expects them to do and how to achieve that. Such budgets tend to be more accurate and closer to the actual situation on the ground.

Types of budgeting

1. Incremental budgeting

Incremental budgeting takes last year's actual figures and adds or subtracts a percentage to obtain the current year's budget. It is the most common method of budgeting because it is simple and easy to understand. Incremental budgeting is appropriate to use if the primary cost drivers do not change from year to year.

2. Activity-based budgeting

Activity-based budgeting is a top-down budgeting approach that determines the amount of inputs required to support the targets or outputs set by the company. For example, a company sets an output target of $100 million in revenues. The company will need to first determine the activities that need to be undertaken to meet the sales target, and then find out the costs of carrying out these activities.

3. Value proposition budgeting

In value proposition budgeting, the budgeter considers the following questions:

- Why is this amount included in the budget?
- Does the item create value for customers, staff, or other stakeholders?
- Does the value of the item outweigh its cost? If not, then is there another reason why the cost is justified?

Value proposition budgeting is really a mindset about making sure that everything that is included in the budget delivers value for the business. Value proposition budgeting aims to avoid unnecessary expenditures – although it is not as precisely aimed at that goal as our final budgeting option, zero-based budgeting.

4. Zero-based budgeting

As one of the most commonly used budgeting methods, zero-based budgeting starts with the assumption that all department budgets are zero and must be rebuilt from scratch. Managers must be able to justify every single expense. No expenditures are automatically "okayed". Zero-based budgeting is very tight, aiming to avoid any and all expenditures that are not considered absolutely essential to the company's successful (profitable) operation.

The zero-based approach is good to use when there is an urgent need for cost containment, for example, in a situation where a company is going through a financial restructuring or a major economic or market downturn that requires it to reduce the budget dramatically.

There are two kinds of the budget in cost accounting that differ in scope, nature, and usefulness.

- A **fixed budget** is a kind of budget where the income and the expenditure are pre-determined. Irrespective of any fluctuation or change, this **budget is static**. Companies that are static, execute the same sort of transactions can significantly benefit from a fixed budget. But wherever there are fluctuations, a fixed budget doesn't turn out to be the most suited one.

- **Flexible budget**, on the other hand, is a budget that is flexible as per the needs of the hour. For example, if the company sees that it can sell off more of its products by expending more in advertisement costs, a flexible budget would help execute that. That's why **a flexible budget is very effective** for companies who go through a lot of changes during a particular period. It is much more complicated than the fixed budget too.

CAPITAL BUDGETING

Capital budgeting refers to the decision-making process that companies follow with regard to which capital-intensive projects they should pursue. Such capital-intensive projects could be anything from opening a new factory to a significant workforce expansion, entering a new market, or the research and development of new products.

Capital budgeting decisions are based on incremental after-tax cash flows discounted at the opportunity cost of capital. Assumptions of capital budgeting are:

- Capital budgeting decisions must be made on cash flows, not accounting income. Accounting profits only measures the return on the invested capital. Accounting income calculations reflect non-cash items and ignore the time value of money. They are important for some purposes, but for capital budgeting, cash flows are what are relevant.
- Cash flow timing is critical because money is worth more the sooner you get it. Also, firms must have adequate cash flow to meet maturing obligations.
- The opportunity cost should be charged against a project. Remember that just because something is on hand does not mean it's free. See below for the definition of opportunity cost.
- Expected future cash flows must be measured on an after-tax basis. The firm's wealth depends on its usable after-tax funds.

Important Capital Budgeting Concepts

Sunk Costs

A sunk cost is a cost that has already occurred and cannot be recovered by any means. Sunk costs are independent of any event and should not be considered when making investment or project decisions. Only relevant costs (costs that relate to a specific decision and will change depending on that decision) should be considered when making such decisions.

A few examples -

- A company spends $5 million on building an airplane. Prior to completion, the managers realize that there is no demand for the airplane. The aviation industry has evolved and airlines demand a different type of plane. The company has a choice: finish the plane for another $1 million or build the new in-demand airplane for $4 million. In this scenario, the $5 million already spent on the old plane is a sunk cost. It should not affect the decision and the only relevant cost is the $4 million.
- A company spends $10,000 training its employees to use a new ERP system. The software turns out to be heavily confusing and unreliable. The senior management team wants to discontinue the use of the new ERP system. The $10,000 spent to train employees is a sunk cost and should not be considered in the decision of discontinuing the new ERP system.
- A company spends $10 million to conduct a marketing study to determine the profitability of a new product they will launch in the marketplace. The study concludes that the product will be heavily unsuccessful and unprofitable. Therefore, the $10 million is a sunk cost. The company should not continue with the product launch and the initial marketing study investment should not be considered when making decisions.

Incremental Cash Flows

Incremental cash flow is the net cash flow attributable to an investment project. It represents the change in the firm's total cash flow that occurs as a direct result of accepting the project.

- Forget sunk costs.
- Subtract opportunity costs.
- Consider side effects on other parts of the firm: externalities and cannibalization.
- Recognize the investment and recovery of net working capital

Independent versus mutually exclusive projects

Mutually exclusive are investments that compete in some way for a company's resources - a firm can select one or another but not both. Independent projects, on the other hand, do not compete with the firm's resources. A company can select one or the other or both, so long as minimum profitability thresholds are met.

When a firm is embarking upon a project, it needs tools to assist in making the decision of whether to invest in the project or not. In order to demonstrate the use of these four methods, the cash flows presented below will be used.

Unlimited Funds vs. Capital Rationing

If a firm has unlimited access to capital, the firm can undertake all projects with expected returns that exceed the cost of capital. Many firms have constraints on the amount of capital they can raise and must use capital rationing. If a firm's profitable project opportunities exceed the amount of funds available, the firm must ration, or prioritize, its capital expenditures with the goal of achieving the maximum increase in value for shareholders given its available capital.

Net Present Value

This method discounts all cash flows (including both inflows and outflows) at the project's cost of capital and then sums those cash flows. The project is accepted if the NPV is positive.

$$NPV = \sum \frac{CF_t}{(1+k)^t}$$

Where CF_t is the expected cash flow at period t, k is the project's where CFT is the expected cash flow at period t, k is the project's cost of capital and n is its life.

- Cash outflows are treated as negative cash flows since they represent expenditure that the company has to incur to fund the project.
- Cash inflows are treated as positive cash flows since they represent money being brought into the company.

The NPV represents the amount of present-value cash flows that a project can generate after repaying the invested capital (project cost) and the required rate of return on that capital. An NPV of zero signifies that the project's cash flows are just sufficient to repay the invested capital and to provide the required rate of return on that capital. If a firm takes on a project with a positive NPV, the position of the stockholders is improved.

Decision rules

- The higher the NPV, the better.
- Reject if NPV is less than or equal to 0

	Cashflows (CF_t)	
Year (t)	**Project A**	**Project B**
0	-5,000	-5,000
1	750	500
2	2,000	1,500
3	2,000	2,000
4	2,500	3,000

Assuming the cost of capital for the firm is 10%, calculate each cash flow by dividing the cash flow by (1 + k)^t where k is the cost of capital and t is the year number. Calculate the NPV for Project A and B above.

NPV = $CF_0 + CF_1 + CF_2 + CF_3 + CF_4$

- Project A's NPV = $-5{,}000 + \frac{750}{1.1} + \frac{2000}{1.1^2} + \frac{2000}{1.1^3} + \frac{2500}{1.1^4} = \$549$
- Project B's NPV = $-5{,}000 + \frac{500}{1.1} + \frac{1500}{1.1^2} + \frac{2000}{1.1^3} + \frac{3000}{1.1^4} = \$246$

Internal Rate of Return (IRR)

It is the discount rate that forces a project's NPV to equal to zero.

$$NPV = \sum \frac{CFt}{(1+IRR)^t} = 0$$

Note this formula is simply the NPV formula solved for the particular discount rate that forces the NPV to equal zero. The IRR on a project is its expected rate of return. The NPV and IRR methods will usually lead to the same accept or reject decisions.

Decision rules

- The higher the IRR, the better.
- Define the hurdle rate, which typically is the cost of capital.
- Reject if IRR is less than or equal to the hurdle rate.

	Cashflows (CF_t)	
Year (t)	**Project A**	**Project B**
0	-5,000	-5,000
1	750	500
2	2,000	1,500
3	2,000	2,000
4	2,500	3,000

- Project A's IRR can be calculated by solving the following equation –
  $-5{,}000 + \frac{750}{(1+IRR)} + \frac{2000}{(1+IRR)^2} + \frac{2000}{(1+IRR)^3} + \frac{2500}{(1+IRR)^4} = 0$

- Project B's IRR can be calculated by solving the following equation –
  $-5{,}000 + \frac{500}{(1+IRR)} + \frac{1500}{(1+IRR)^2} + \frac{2000}{(1+IRR)^3} + \frac{3000}{(1+IRR)^4} = 0$

Since it is difficult to determine the IRR by hand, the use of a financial calculator is needed to solve for IRR.

The IRR for Project A is 14% and for Project B is 12%

Payback Period

It is the expected number of years required to recover the original investment. Payback occurs when the cumulative net cash flow equals 0.

	Project A		Project B	
Year (t)	**CF**	**Cumulative CF**	**CF**	**Cumulative CF**
0	(1,000)	(1,000)	(1,000)	(1,000)
1	750	(250)	100	(900)
2	350	100	250	(650)
3	150	250	450	(200)
4	50	300	750	550

We see that in case of project A, the cumulative cashflows reach 0 between time 1 and 2. Thus, by simple interpolation,

$$\frac{PP - 1}{2 - 1} = \frac{0 - (-250)}{100 - (-250)}$$

Solving the equation, payback period (PP) for Project A = 1.71 years

Similarly, payback period for project B is 3.27 years

<u>Decision rules</u>

- The shorter the payback period, the better.
- A firm should establish a benchmark payback period. Reject if payback is greater than benchmark

<u>Drawbacks</u>

- It ignores cash flows beyond the payback period. Payback period is a type of "break even" analysis: it cares about how quickly you can make your money to recover the initial investment, not how much money you can make during the life of the project.
- It does not consider the time value of money. Therefore, the cost of capital is not reflected in the cash flows or calculations.

Discounted Payback Period

It is similar to the regular payback method except that it discounts cash flows at the project's cost of capital. It considers the time value of money, but it ignores cash flows beyond the payback period.

	Project A			Project B		
Year (t)	**CF**	**Discounted CF**	**Discounted Cum. CF**	**CF**	**Discounted CF**	**Discounted Cum. CF**
0	(1,000)	(1,000)	(1,000)	(1,000)	(1,000)	(1,000)
1	750	750/1.1 = 681	(319)	100	100/1.1 = 91	(909)
2	350	$350/1.1^2 = 289$	(30)	250	$250/1.1^2 = 207$	(702)
3	150	$150/1.1^3 = 113$	83	450	$450/1.1^3 = 338$	(364)
4	50	$50/1.1^4 = 34$	117	750	$750/1.1^4 = 512$	148

We see that in case of project A, the discounted cumulative cashflows reach 0 between time 2 and 3. Thus, by simple interpolation,

$$\frac{DPP - 2}{3 - 2} = \frac{0 - (-30)}{83 - (-30)}$$

Solving the equation, discounted payback period (DPP) for Project A = 2.26 years

Similarly, payback period for project B is 3.71 years

Average Accounting Rate of Return

ARR = Average Annual Profit/ Average Investment

Where,

- ***Average Annual Profit = Total profit over Investment Period / Number of Years***
- ***Average Investment = (Book Value at Year 1 + Book Value at End of Useful Life) / 2***

Drawbacks

- It does not take into account the time value of money - the value of cash flows does not diminish with time as is the case with NPV and IRR.
- ARR is based on numbers that include non-cash items

Profitability Index

This is an index used to evaluate proposals for which net present values have been determined. The profitability index is determined by dividing the present value of each proposal by its initial investment.

PI = PV of future cash flows / Initial investment = 1 + (NPV / Initial investment)

The PI indicates the value you are receiving in exchange for one unit of currency invested.

- An index value greater than 1.0 is acceptable and the higher the number, the more financially attractive the proposal.
- A ratio of 1.0 is logically the lowest acceptable measure on the index. Any value lower than 1.0 would indicate that the project's PV is less than the initial investment.

NPV vs IRR

The Relative Advantages and Disadvantages of the NPV and IRR Methods

A key advantage of NPV is that it is a direct measure of the expected increase in the value of the firm. NPV is theoretically the best method. Its main weakness is that it does not include any consideration of the size of the project. For example, an NPV of $100 is great for a project costing $100 but not so great for a project costing $1 million.

A key advantage of IRR is that it measures profitability as a percentage, showing the return on each dollar invested. The IRR provides information on the margin of safety that the NPV does not. From the IRR, we can tell how much below the IRR (estimated return) the actual project return could fall, in percentage terms, before the project becomes uneconomic (has a negative NPV).

The disadvantages of the IRR method are (1) the possibility of producing rankings of mutually exclusive projects different from those from NPV analysis and (2) the possibility that a project has multiple IRRs or no IRR. (3) There is an implicit assumption that the cashflows will be reinvested at the IRR.

Similarities of Outcomes under NPV vs IRR

Both methods show comparable results regarding “accept or reject” decisions where independent investment project proposals are concerned. In this case, the two proposals don’t compete, and they are accepted or rejected based on the minimum rate of return on the market.

Conventional proposals often involve a cash outflow during the initial stage and are usually followed by a number of cash inflows. Such similarities arise during the process of decision-making. With NPV, proposals are usually accepted if they have a net positive value, while IRR is often accepted if the resulting IRR has a higher value compared to the existing cut off rate. Projects with a positive net present value also show a higher internal rate of return greater than the base value.

Conflicts Between NPV vs IRR

In the case of mutually exclusive projects that are competing such that acceptance of either blocks acceptance of the remaining one, NPV and IRR often give contradicting results. NPV may lead the project manager or the engineer to accept one project proposal while the internal rate of return may show the other as the most favourable. Such a kind of conflict arises due to a number of problems.

For one, conflicting results arise because of substantial differences in the amount of capital outlay of the project proposals under evaluation. Sometimes, the conflict arises due to **issues of differences in cash flow timing and patterns** of the project proposals or **differences in the expected service period** of the proposed projects.

When faced by difficult situations and a choice must be made between two competing projects, it is best to choose a project with a larger positive net value by using cut-off rate or a fitting cost of capital.

The reason the two above mentioned options work is because a company's objective is maximizing its shareholder's wealth, and the best way to do that is choosing a project that comes with the highest net present value. Such a project exerts a positive effect on the price of shares and the wealth of shareholders.

So, NPV is much more reliable when compared to IRR and is the best approach when ranking projects that are mutually exclusive. Actually, NPV is considered the best criterion when ranking investments.

Also, it can be demonstrated that the better assumption is the cost of capital for the reinvestment rate Multiple IRRs is the situation where a project has two or more IRRs. This problem is caused by non-conventional cash flows of a project.

- **Conventional cash flows** means that the initial cash outflows are followed by a series of cash inflows.
- **Non-conventional cash flows** means that a project calls for a larger cash outflow either sometime during or at the end of its life. Thus, the signs of the net cash flows flip-flop during the project's life.

In fact, non-conventional cash flows can cause other problems such as negative IRR or an IRR, which leads to an incorrect accept or reject decision. However, a project can have only one NPV regardless of its cash flow patterns so the NPV method is preferable when evaluating projects with non-normal cash flows.